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Relativity and the Status of Becoming

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The merging of space and time proposed by Minkowski in 1908 is still sometimes misinterpreted as a sort of four-dimensional hyperspace of which time is the fourth dimension, analogous to the other, spatial dimensions. An inevitable consequence of this view is that the future events somehow exist prior to, and independently of, human awareness and that what we call "becoming" is "merely a coming into our awareness" (A. Grünbaum). However, an attentive inspection of the space–time diagram and of Minkowski's formula for the constancy of the world interval shows that the events contained in the absolute future of any frame of reference are intrinsically unobservable *not only within this system, but also by any other conceivable observer: consequently, there is no reason to postulate their existence.*

1. THE STRUCTURE OF MINKOWSKI'S TIME–SPACE

Since the coming of relativity theory it has been repeatedly asserted that Minkowski's concept of space–time is incompatible with the objective status of becoming in the physical world. Although this view has been repeatedly and effectively criticized by various authors, it crops up again and again, in most instances based on the same old fallacious arguments and disregarding the objections which had been already raised against them. This view is often phrased differently, but its import remains the same. Thus

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Williams,[1] and, more recently, Park,[2] speak of "the myth of the passage"; Quine says that the discovery of the principle of relativity "leaves no reasonable alternative to treating time as spacelike"[3]; according to Grünbaum, the same principle implies that "the coming into being is only *coming into present awareness*" (author's italics)[4][1]; finally, in Putnam's less unambiguous words, "we must say that future things are real, even if they do not exist yet."[5] Illustrations of this sort could be multiplied, even though the representatives of the "becomingless view" are now fewer than in the years immediately following Minkowski's formulation of the concept of space–time.

Interestingly, among those who uphold the becomingless view there are some significant differences. In the first place, those who adhered to this view only occasionally while expressing an opposite view at other times should be properly excluded from this group. Such was clearly the case of Einstein[6] and Weyl.[7] But equally excluded should be those who, while speaking of Minkowski's four-dimensional continuum, really mean the space–time of classical physics in the sense of Lagrange and d'Alembert, who were the first to call time "the fourth dimension" and mechanics "geometry of four dimensions." Such certainly is the case of Donald Williams, who is clearly unaware of the basic difference between classical and relativistic space–time.

But even among the better-informed defenders of the static interpretation of space–time there are significant differences of view. Grünbaum emphatically insists that the elimination of becoming, allegedly implied by the structure of relativistic space–time, does *not* imply determinism; Putnam (Ref. 5, pp. 243–247) and Rietdijk[8] equally emphatically (and more consistently) claim that it does. In view of the fact that Grünbaum shares with Putnam the belief in the attribute-specificity of the future events, one would tend to dismiss the above-mentioned disagreement as merely of semantic kind. Yet, if it is so, Grünbaum is unaware of it; he repeatedly insists that "the issue determinism and indeterminism is *totally irrelevant* to whether becoming is a significant attribute of the time of physical nature independently of human consciousness" (Ref. 4, p. 321). Thus the difference between those two views is very real; I shall return to it in the concluding part of this article.

The most frequent and most plausible argument for the static, i.e., becomingless, interpretation of Minkowski's space–time is the relativization of simultaneity. Its plausibility is understandable: If there is no objective cosmic "Now" in the sense of being invariant for all frames of reference, then the distinction between the past and the future seems to be lost. In Newtonian space–time (the word itself was not used, but its meaning was one of the cornerstones of classical physics) it was possible to make at each

[1] See Ref. 7 for an answer to Ref. 4.

particular instant an *instantaneous cut* separating unambiguously the past from the present. Each such instantaneous cut represented a three-dimensional layer of *objectively simultaneous events*, that is, an instantaneous space with its physical content at that moment. But since no such instantaneous cuts, no such "world-wide instants" invariant for every observer are possible in a relativistic space–time, there is no absolute "Now," and consequently no absolute separation between the past and the future: The distinction between the successive states of the universe seems to be obliterated and we thus would have Putnam's conclusion that "future things must be real." Hence there would be no becoming in the physical world.

A more attentive analysis of the structure of relativistic space–time will show that this is a superficial and unwarranted conclusion and that the relativity theory *strengthened* rather than weakened the objective status of becoming. In the first place, there are some types of succession which are *absolute* since they remain successions in *all* frames of reference. This is true of all causally related events whose temporal order is *topologically invariant* in all possible frames of reference. More specifically: The succession of the events whose spatial separation is smaller than their separation in time multiplied by the velocity of electromagnetic waves can never degenerate into simultaneity by *any* choice of the frame of reference; a fortiori, it can never be inverted. In this respect the temporal order of such events is *fundamentally different* from what Reichenbach called "unreal temporal sequences" (*die irreelle Zeitfolgen*)[9] of causally unrelated events. Such sequences are *not* topologically invariant since they can be inverted by an appropriate change of the frame of reference. Reichenbach's term is very appropriate because it correctly indicates the fictitious character of such "sequences." For the relativistic space–time—whose more appropriate name should be *time–space*—consists of a network of world lines each of which consists of a succession of irreversible, causally related events; but no world line, no concrete physical connection corresponds to such "unreal sequences"; their status is *purely conceptual*, comparable to the status of "simultaneity of distant events."

The failure to stress the second, and, in my view, the most significant, aspect of the relativistic time–space is due to the fact that the phrase "relativization of simultaneity" does not immediately suggest the implication of this concept for space. It is certainly surprising how often it is overlooked that the three terms— "the class of absolutely simultaneous events," "absolute instantaneous space," and "absolute juxtaposition"—are entirely synonymous; yet they are three different terms designating the same concept. The negation of absolute simultaneity is implied by the negation of instantaneous space and vice versa. Since there is no absolute simultaneity of distant events, *there is no absolute juxtaposition either*. Thus while there are

no juxtapositions (i.e., instantaneous spatial relations) which would be juxtapositions in all frames of reference, *there are certain types of succession which remain such in all frames of reference.*

All this follows inescapably from Minkowski's formula for the invariance of the world interval and it would be otiose to restate it again in an explicit mathematical form.[2] These implications are incompatible with the tendency to regard Minkowski's fusion of space and time as a sort of *four-dimensional hyperspace*, time being its additional and equally static dimension. Putnam and Quine do it quite explicitly and so does Grünbaum: What else is his theory of "time dependence of becoming"? Unfortunately, a subtly static connotation of Minkowski's original term "the four-dimensional *world*" contributed to this confusion. I think that an adequate characterization of Minkowski's time–space is as follows: There is a four-dimensional process of gradually unfolding causal lines separated not by instantaneous distances, but by four-dimensional loopholes of causal independence.

2. THE PHYSICAL EMPTINESS OF THE FUTURE

If no empirically verifiable physical reality corresponds to "absolute simultaneity" and instantaneous "spacelike connections," the same is equally true of the future events. By "future events" I do not mean those *nominally* future events whose relations to my Here–Now are temporally indeterminate since they belong neither to my absolute past nor to my absolute future. For such events are merely declared to be future because they apparently lie on the forward side of my "now line" by which I arbitrarily divide my own region of Elsewhere; to another observer with a different velocity and drawing a different "now line" they may appear either simultaneous or in his noncausal past. (The word "appear" is really misleading since all events in "Elsewhere" are not only never perceived, but also *unperceivable in principle.*) Thus by "future" I mean my own *causal* and *absolute* future symbolized in a three-dimensional time–space diagram by the forward causal cone radiating from my Here–Now. I am going to show that such a future is *truly absolute*, that is, intrinsically unperceivable not only by me, but also by any possible observer. By "any possible observer" I mean any frame of reference anywhere in my causal past or in my region of Elsewhere.

The proof of it is simple since it follows from Minkowski's formula for the invariance of the world interval. From it follows not only that the temporal order of causally related events cannot be reversed by any change of the frame of reference, but also that the succession of such events can

[2] As shown for the first time by Langevin.[10]

never degenerate into simultaneity for any observer. Consider any event in my causal future; it is, in principle at least, a causal successor of the event Here–Now and in this sense it will occur *after* my own experienced present. (The presence of my own person or of any conscious percipient is really irrelevant since even a "merely physical" Here–Now will exhibit the same relationship to its own causal future.) This obviously means that not only can I not perceive any event of my future life, nor, physically speaking, any event on the "future part" of my own world line, but neither can I perceive any event *on any other world line which will be included in my causal future.* This is a consequence of the fact that *the necessary condition of the observability of any event is to be included in the causal past of the corresponding observer.* Nothing in my causal future can causally affect my Here–Now, for the simple reason that my own causal past and causal future do not overlap. This conclusion may appear truistic, but considering the persistence of the Wellsian "time-tunnel" phantasies and the speculations about precognition, it is hardly superfluous to stress it.

But perhaps some event in my causal future can be perceived by some other distant observer located in my Elsewhere region? This would mean that such future event would appear to this observer *before* it appears to me. But this would require that he receive a signal from my causal future proceeding with a velocity greater than that of light. The impossibility of this can be seen at once from the relativistic space–time diagram: No part of the forward causal cone symbolizing my absolute future is included in the rearward cone (i.e., causal past) of *any* observer in Elsewhere. (A fortiori, it cannot be perceived by any observer located in my causal past.) The very definition of Elsewhere excludes any possibility of its communication with my causal future.

There is another aspect of the same consequence: The impossibility of the inversion of the temporal order of two causally related events. For the observer in Elsewhere receiving a signal from my absolute future would perceive a future event causally dependent on my present Here–Now *before* he perceives my Here–Now! Clearly, as long as we remain on relativistic grounds, the conclusion is inescapable: The events constituting my absolute future are *unobservable in principle by any conceivable observer.* And since my own frame of reference has no privileged character, we must generalize the result obtained, in the following way: No event future with respect to *any* particular Here–Now can be perceived by *any* then conceivable observer.

In the light of what has been just said it becomes clearer what is either incorrect or confused in the writings of Rietdijk, Putnam, and Grünbaum. Rietdijk's main error was to assume the possibility of instantaneous causal interaction between an event in Elsewhere and Here–Now when he spoke about "experiencing" Here–Now by a distant observer in Elsewhere (Ref. 8,

p. 342). There is, of course, no such experience; "instantaneous telepathy" has no place in relativistic physics, nor is there any place for precognition. Yet, this is what Rietdijk assumes when he claims that the event P in the absolute future of one observer is "already in the past" for another observer moving with respect to the first one (Ref. 8, p. 341). If this statement is to have any operational meaning, the term "past" should mean "causal past"; but then this statement is plainly *false* for the reasons given above. The main source of Rietdijk's confusions is his failure to differentiate between "causal past," which is experienceable in principle, and a merely nominal "noncausal past," which is merely *inferred*—differently by different observers—and never observed or observable. It is also obvious that a similar distinction between causal and noncausal future escapes him. For the "noncausal past" and "noncausal future" are merely two parts of an Elsewhere region separated by a fictitious "Now line"; and since such "Now lines" are drawn differently by different observers, none of them can claim any objective significance for his own distinction between his noncausal past and noncausal future.

This is merely another aspect of the elimination of absolute simultaneity of nonisotopic (i.e., spatially not coinciding) events. I say "elimination"[3] instead of mere "relativization" since the latter, "milder" term tends to be misleading. For the very distinction between noncausal past and noncausal future is based on the unconscious retention of the classical Newtonian "Now" which, grafted incongruously on the relativistic framework, produces the confusions just mentioned. It is a new instance of an "old wine being poured into new vessels," with obvious results. We stress again that the elimination of the distinction between a fictitious noncausal past and an equally fictitious noncausal future in no way affects the objective difference between absolute past and absolute causal future. On the contrary, as Eddington wrote long ago, the past and the future are now separated even *more effectively* than in classical physics; for while in the physics of Newton the past was separated from the future by an infinitely thin layer of simultaneous events constituting the allegedly objective "Now," in Minkowski's time–space it is the whole four-dimensional "cone" of Elsewhere which separates them. Many confusions in interpretations of the relativity theory are due to the failure to distinguish between the genuine (i.e., causal or absolute) and nominal (noncausal) future. Similarly for the past. Apparent paradoxes of the reversal of the temporal order occur only when we deal with the noncausal, i.e., *inauthentic*, past and future.

Much—in truth, nearly all—of what I said about Rietdijk's article can be repeated with respect to Putnam's view. Like Rietdijk, Putnam retains the old notion of the universal present spread as a "world-wide instant"

[3] In Einstein's words: "There is no such thing as simultaneity of distant events."[13]

across the whole universe and uses this notion, in order eventually to conclude that, in a sense, *everything* is present. He disregards one essential idea of relativity, that, as I have said elsewhere,[12] "Here–Now" can never be extrapolated to "Everywhere–Now"; or, in Stein's words, "in Einstein–Minkowski's space–time an event's present is constituted by itself alone. In this theory, therefore, the present tense can *never* be applied to 'foreign' objects."[13] Robb expressed this truth more than fifty years ago when he wrote "that we cannot strictly identify the same instant at two distinct points of space"; in other words, "the present instant, properly speaking, does not extend beyond itself."[14] If our imagination refuses to accept this, it only shows how thoroughly Newtonian it remains. Finally, Putnam's claim that "the future events are real" is based on the same failure to differentiate between absolute, causal future, unobservable by any conceivable observer, and purely nominal noncausal "future" which is, properly speaking, no future at all, but merely a part of the Elsewhere region. But Putnam gives his argument a label which has a plausible relativistic ring, saying that there are no privileged observers. Since Grünbaum uses a similar argument, let us consider it in the form he gave it.

Grünbaum borrowed his argument from Hugo Bergmann, who nearly a half century ago insisted against Reichenbach that "Now" has no objective physical significance, being merely "the temporal mode of the experiencing ego."[15],[4] In other words, all various successive "Nows" are on the same footing, whether it is the "Now" of Plato's birth in 427 B.C. or the present (1975) "Now" of this writing or the "Now" of George Orwell's 1984. Or in Putnam's words: no privileged observer means no privileged "Now." It can be easily shown that (a) this argument has no basis in relativistic physics; (b) it leads to needless complications and unsolvable epistemological puzzles.

Re (a): The Bergmann–Grünbaum argument is as follows: At every particular Here–Now the absolute future is unambiguously separated from its absolute past. But since this is true of any particular Here–Now, it follows that all such events, i.e., all successive "presents," are *entirely equivalent* and none of them has any privileged character; objectively, i.e., from the physical standpoint, they are all on the same level. Thus the only difference between my present (1975) and the 1984 "present" is not physical at all, but only subjective or psychological; the fact that I am experiencing my own Now does not make the past and the future "nows" less real. Consequently, what we call becoming is merely a "coming into our own awareness."

This is Grünbaum's thesis of the "mind-dependence of becoming" and, before pointing out its bizarre epistemological consequences, let us first see why it does *not* follow from the relativistic structure of time–space. For there

[4] Also see Grünbaum,[4] pp. 27–28.

is a *physically observable difference* between Plato's "now" and the "now" of Orwell's novel; the former is included not only in my own causal past, but also in that of countless other observers—in truth, in the causal past of all frames of reference which are within the forward light cone whose vertex was on the earth (more accurately in Athens) in 427 B.C. On the other hand, the events of 1984 are not included in the causal cone of any conceivable observer. In a simple language: This is why we can read Plato's works while we cannot yet read what Grünbaum will write ten years from now. When it is claimed that these future events are "tenselessly" included in the causal past of the observers *posterior* to 1984, we merely translate into a strangely tenseless language the obvious fact that any future event *will be* eventually past for our still more remote future descendants. But nothing is thus gained since our remote future descendants are as unreal and as *intrinsically unobservable* as the future year 1984 itself! Why complicate our physics by postulating such intrinsically unobservable entities as future events?

Re (b): This complication becomes even more startling when we consider its epistemological aspect. For the theory of "mind dependence of becoming" in calling "now" merely "the temporal mode of experiencing ego" excludes becoming from the physical realm while retaining it on the psychological level. But in this way a paradoxical and intolerable dualism of two completely heterogeneous realms is created and there is no way in which these two realms can be intelligibly related. No semantic fog and no convolutions of style can successfully hide this difficulty. It is the worst form of "bifurcation of reality," which in its sharpness surpasses the dualism of Descartes; for no matter how distinct the two Cartesian substances were, they at least were *both in time*, i.e., subject to change, i.e., to becoming. Grünbaum's answer to this is his fictitious distinction between "time" and "becoming"; he says that while the physical events are devoid of becoming, they still remain *temporally separated.* Note the characteristic word "separation": A little effort of introspective analysis should convince him that what he has in mind when he speaks of "temporal separation of events" is merely a *juxtaposition* of distinct points *coexisting* on an already drawn line which he still calls "time." Two successive events are thus represented by two points A and B and their distance $\overline{AB}$ represents their temporal "separation"! Obviously, on a static geometrical line there is no place for becoming or genuine succession; what remains is a static symbol which he still labels "time" and the *coexisting* events which he calls "successive." Only thus can Grünbaum's strange view of becomingless time become *psychologically* understandable without becoming more satisfactory; it is based on a common confusion of time itself with its geometrical symbol, the confusion denounced by Bergson long ago. He is indignant when I characterized his view of physical reality as a "bloc universe"; yet, is there any other *more* appropriate

term to describe his becomingless physical world? In his words "coming into being is only *coming into present awareness*"; this means, if anything, that what I experience as a new sensory or introspective quality existed timelessly (Grünbaum prefers to say: "tenselessly") *prior to* and *independently of* my present awareness. It is difficult to be more explicit and closer to the neo-Eleatic metaphysics of "the bloc universe."

There is an additional difficulty here. It is rather strange to find both Putnam and Grünbaum, who are both physicalistically oriented, in the company of such "timeless" idealists as Bradley and McTaggart. Apparently, the common aversion to becoming produces some strange alliances. Such metaphysics, which regards all successive "nows" as equally real, faces embarrassing questions: "Why are we not living in the reign of George III? Or in the Cretacious period? Or in 1984?" The answer of the "timeless" idealists to this question is based on their distinction between "appearance" and "reality": Becoming belongs to the former, not to the latter. It is a very strange dualism, but at least in F. H. Bradley's philosophy becoming had a certain *locus* since it belonged to the subjective realm. But the physicalists cannot consistently use this device since in their view "mind," including the subjective present awareness, is just another part of the physical and (in their view) becomingless world. Thus the physicalistic neo-Eleatics are in a far more difficult position since their own view makes even the *appearance* or *illusion* of becoming impossible. The questions above are then really unanswerable.

Elimination of becoming implies rigorous determinism; unlike Grünbaum, both Putnam and Rietdijk were consistent in accepting this conclusion. But since the premise is false, the conclusion does not follow. It is usually said that relativistic physics remains deterministic because of its *macroscopic* character. This is not entirely true; the most spectacular verifications of the special theory of relativity had been made in microphysics (the increase of mass with velocity, the mass defect, the time dilation in meson decay, etc.). Any satisfactory discussion of the problem of determinism in relativity has to take into account the problem of causality in microphysics and the relation of microphysical indeterminacy to the realm of the middle dimensions.[5] Such discussion is clearly beyond the scope of this article.

3. SUMMARY

We thus arrive at the following conclusions.

(a) The view that the physical reality is devoid of becoming implies that "becoming" of events is merely "their coming into our present

[5] On this problem see Margenau.[16]

awareness"; in other words, that the future physical events exist *prior to* and *independently of* our present awareness.

(b) Nothing in relativistic physics supports this view; on the contrary, at any particular moment, future events are *intrinsically unobservable* by any conceivable observer.

(c) If we persist in postulating the actual preexistence of future events in spite of their intrinsic unobservability, we violate the most elementary rule of scientific methodology; instead, we should place such intrinsically unobservable events into the same category as phlogiston, caloric, motionless aether, and other useless and discredited fictions.

(d) Furthermore, the denial of becoming in the physical world generates enormous epistemological difficulties in creating an intolerable dualism of two completely heterogeneous realities without any possibility to relate them in any intelligible way: the becomingless realm of "true" reality and the temporally incomplete realm of the phenomenal "flux"; basically it is a relapse into ancient Eleatic myths with all their weird and superfluous paradoxes.

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